

The Mouse Race

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Executive Summary

This project focuses on the development of an autonomous, battery-powered mouse system capable of completing a 20 m wire-guided track by detecting a 20 kHz electromagnetic signal. The objective was to produce a reliable, efficient system that controls speed, stability, and energy consumption within a 1-week project timeframe. The team had to integrate sensing, control, power electronics, and mechanical systems for the mouse to consistently lap the circuit in a reasonable time.

Multidisciplinary methods were used to develop the system. We used a pick-up coil to detect the electromagnetic field created by the track wire, producing a low-amplitude signal. This signal was processed using a passive band-pass filter and peak detector to isolate the 20 kHz frequency and reduce noise. An Arduino was used to generate PWM signals from this signal to control motor speed using a MOSFET-based motor driver. We used LTspice simulations to validate circuit designs before implementing them, ensuring they behaved correctly to reduce hardware iteration time. The team conducted tests using oscilloscopes and waveform generators to verify signal integrity, PWM behaviour, and motor response. Chassis design considerations also ensured that the sensors were stable and motor torque was transferred efficiently.

The results showed that the system successfully detected the track signal and enabled controlled motor operation. PWM control significantly improved efficiency compared to linear methods by reducing power dissipation. Signal filtering improved the accuracy of the detected signal, allowing smoother steering and reducing unnecessary corrections. The mouse could follow the track partially during testing, requiring us to adjust its position every turn, but a perfect lap without readjustment would require more optimisation and system corrections.

In conclusion, the project part-successfully demonstrated the integration of sensing, signal processing, and motor control in a functional autonomous system. Design simulations and iterative testing were essential in achieving reliable operation. There were some key issues yet to be solved as the final system could not lap the track without help, suggesting areas for further improvement.

The project significantly impacted our development of practical engineering skills, including circuit design, simulation, testing, and system integration. It also reinforced the importance of structured teamwork, clear communication, and problem-solving in delivering a complex engineering system under tight time constraints.

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List of Acronyms

UoB	University of Bath
PWM	Pulse Width Modulation
D&D	Design and Development
LCA	Life Cycle Assessment
SMART	Specific Measurable Achievable Relevant Time-bound

1. Introduction

1.1 Aims

The main aim of this project week is to have a function mouse system ready for the final race, and it has been documented using the SMART acronym below.

Specific: Design and develop an autonomous mouse robot capable of detecting a 20 kHz track signal and following a 20 m wire path.

Measurable: Achieve stable signal detection and motor control, with successful partial or full track completion during testing and a target wheel speed of approximately 400 - 430 rpm.

Achievable: Implement the system using the given components (NiMH AA / PP3 batteries, Arduino microcontroller, MOSFET motor driver, and filters) within the lab sessions.

Relevant: Integrate sensing, control, and power electronics to demonstrate practical engineering design and system-level problem solving.

Time-bound: Complete design, simulation, assembly, and validation within the project week, with a functional prototype ready for final test.

1.2 Scope

This project covers the design, simulation, construction, and testing of an autonomous mouse robot, including signal detection using a pick-up coil, signal conditioning via filtering and peak detection, microcontroller-based control, and PWM motor driving. It also includes performance testing, optimisation, and sustainability considerations such as energy use and material management. Advanced features such as fully optimised closed-loop control or extended long-term durability testing are beyond the scope of this project.

2 Team management

2.1 Team leadership

Team charter

Team Purpose / Mission

The purpose of our team is to design, build and test a battery powered autonomous “mouse” capable of completing the 20 m wire track as quickly and reliably as possible during the project week. The project required integrating sensing, control, electronics and mechanical design to produce a functional prototype capable of following the

magnetic field generated by the track wire. We aimed to develop a design that completes laps consistently while balancing speed, stability and power efficiency.

Goals and Deliverables

Our primary goal was to produce a working autonomous mouse capable of completing the track reliably within the timed conditions. Success was defined by the mouse completing the full track consistently and achieving competitive lap times during our testing.

Key deliverables included:

- A functional mouse prototype capable of detecting the track signal.
- Integration of sensing, steering and motor systems.
- Testing and optimisation on practice tracks.
- A documented project report describing the design, testing and results.

3 key SMART goals include:

1. Filter Circuit and Peak Detector

Design and implement a passive band-pass filter with a peak detector that isolates the 20 kHz signal from the pick-up coil and reduces unwanted noise. The circuit will be tested using an oscilloscope to confirm that the output signal amplitude is stable and significantly cleaner than the raw coil signal. The goal is to achieve a clearly measurable output suitable for microcontroller input by the end of the electronics testing phase within the next two weeks.

2. Motor Drive Circuit

Design, build, and test a MOSFET-based motor drive circuit capable of controlling the speed of the DC motors using Arduino PWM signals. The circuit should allow stable motor operation and adjustable speed control, verified through testing and measurement of wheel RPM. The target is to achieve reliable PWM control and a wheel speed of approximately 430 rpm before the system integration stage within two weeks.

3. Mechanical Integration

Integrate the electrical components, including the motor drive circuit, sensors, and microcontroller, into the mechanical chassis of the mouse robot. The assembly must securely mount the motors, wheels, and pick-up coil while maintaining proper alignment with the track wire. The goal is to complete the mechanical assembly and ensure all systems operate together successfully by the final race phase of the project.

Roles and Responsibilities

To keep the work organised responsibilities were shared across the team based on individual strengths and interests. Although we worked collaboratively throughout the project each team member took the lead on a specific area while still supporting others when needed. This helped ensure that all parts of the system were covered while maintaining shared responsibility for the overall project. Each area had a designated lead to maintain accountability and help coordinate work within that part of the project:

Role Area	Assigned "Lead"
Sensor system design	msmk23
Motor driver electronics	jh4053
Programming / control	mua35
Mechanical integration	yamae20
Testing and optimisation	xm482
Documentation/or ganisation	sk3307

Figure 1: Designated roles

Ways of Working / Team Processes

The team held regular discussions throughout the project week to review, progress, coordinate tasks and plan testing sessions. Communication took place through group messaging and in person discussions during our scheduled lab sessions. We also created a Microsoft team to communicate efficiently with each other.

Design decisions were made through group discussion using testing results and technical reasoning. Work was divided so that different parts of the system such as mechanical assembly, electronics and testing could be carried out at the same time. This helped us make better use of the available lab time and move the process along much more efficiently.

Norms and Behavioural Expectations

The team aimed to maintain a respectful and supportive working environment. Everyone was encouraged to share ideas and take part in discussions regardless of their role. Constructive feedback was encouraged and disagreements were handled respectfully with the aim of improving the design not criticising individual's ideas. Everyone was prompted to arrive at the labs by 10am so that we could crack on with the hefty amount of work that had to be done.

Team members were expected to contribute actively communicate clearly and support each other when problems came up during the design and testing stages.

Skills matrix

Team Member		Sensor System Design	Motor Driver Electronics	Programming / Control	Mechanical Integration	Testing & Optimisation	Documentation & organisation
shah	sk3307	3	2	2	4	4	5
yassin	yamae20	2	2	3	3	3	3
markus	msmk23	3	3	3	4	4	2
usman	mua35	2	3	3	4	4	5
xavier	xm482	2	2	2	3	3	4
jaap	jh4053	3	3	3	4	4	4

Figure 2: Skills matrix

Decision-making process

If there were technical decisions to be made, there would be a majority vote with the team lead. Ultimately it would be the decision of the team lead to make but general votes would help determine what the best decision to make would be.

No emergency or budget decisions had to be made in our case, as our components were being funded by the university.

Process for Controversy with Civility

Our team used a structured discussion approach when different design ideas came up. Each person explained their idea and reasoning before the group discussed it. Everyone had the chance to speak without interruption and team members were encouraged to listen carefully to each other. After discussion we tried to reach a group agreement. If this was not possible the team decided through a majority decision. This helped keep discussions respectful while still allowing everyone's different ideas to be explored.

Impact of the Process

This approach made it easier for everyone in the team to share their ideas and opinions. Because discussions were structured misunderstandings were less likely, and disagreements stayed focused on the design rather than becoming personal. As a result, decisions were made based on testing results and technical reasoning rather than individual preference. This improved teamwork and helped us consider different solutions when developing the autonomous mouse and helped avoid unnecessary conflict within the team.

Resources and constraints

We used many different tools for different purposes during the week. We utilised WhatsApp and Microsoft teams for effective communication within our group so that ideas and flow of work could be transparent to everyone. We made use of LTspice for

circuit simulation. We simulated the motor drive circuit before actually making it on breadboard so that we had a brief idea of what we were making, and we decided to simulate the switch-mode circuit for the motor drive. We also simulated the filter circuit to similarly get an idea of how the filter system would work.

There was no such budget during this project week as it was all funded by the university, but we still ensured that we weren't being wasteful and using the equipment provided to us sensibly.

The only main constraint that we had was that the labs are always set to close at 5pm, which limits the amount of work that can be done in a day. This meant that all the work that we had to do and test had to be done before 5pm sharp, or else we would go into the next day with more tasks to be done.

Team values

There are many different principles that guided our teamwork, and these principles governed how well we cooperated with one another, and it prevented arguments from breaking out. These include:

- Accountability
- Respect for differing opinions
- Continuous improvement
- Controversy with civility

3 Mouse Design

3.1 Dynamic system capture and simulation using SIMULINK

The dynamic behaviour of the mouse robot was analysed by modelling key subsystems, including the motor drive and signal conditioning circuits. The system can be represented as many energy conversions and control processes, where a low-amplitude input signal from the pick-up coil is conditioned and used to regulate motor speed.

The motor system was modelled as a DC motor driven by a PWM-controlled voltage source. The PWM signal represents a switching input, where the duty cycle determines the average voltage applied to the motor and therefore its speed. This enables dynamic control of the system, as variations in the input signal directly influence motor response.

For signal conditioning, we used a band-pass filter followed by a peak detector and smoothing stage. The filter isolates the 20 kHz track signal while attenuating noise, ensuring that the system responds only to the intended input. The peak detector and smoothing capacitor convert the high-frequency AC signal into a stable DC voltage proportional to signal strength, which forms the control input for the system.

By combining these models, the system behaviour can be understood as a feedback-driven process where signal strength influences motor control. Simulation allows the dynamic response of the system to be analysed before implementation, ensuring stability, responsiveness, and correct signal interpretation.

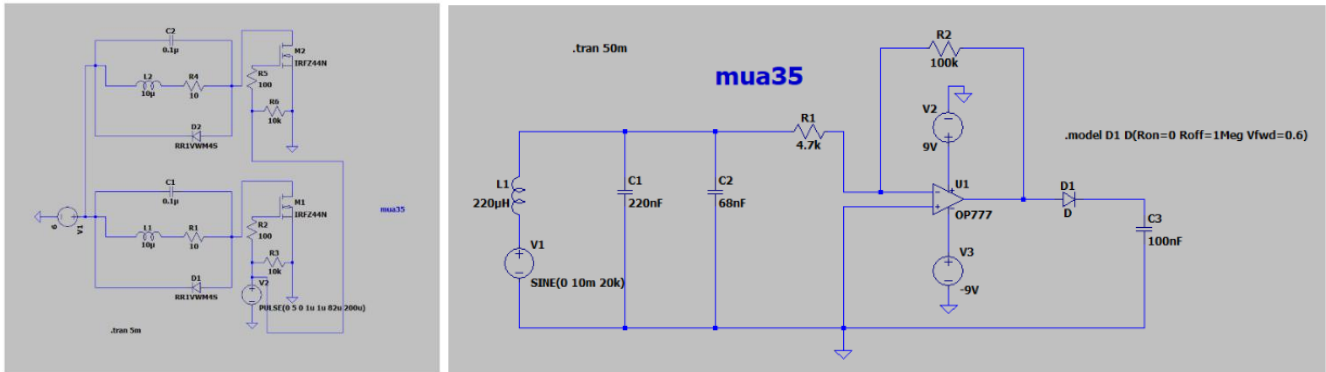


Figure 3: Motor drive and peak detector circuit simulations

3.2 Power electronics simulation and design evaluation

The power electronics design was evaluated using LTSpice to verify the operation of both the motor drive and signal conditioning circuits before implementation. This simulation-based approach reduced the need for repeated hardware testing and improved overall reliability.

The motor drive circuit utilises a MOSFET-based switching system to implement PWM control. Simulation results showed that the MOSFET operates as a high-speed switch, producing a square-wave output corresponding to the PWM input signal [16]. The duty cycle of this waveform directly controls the average voltage supplied to the motor, allowing efficient speed regulation. Compared to linear control methods, this switching approach significantly reduces power dissipation as heat, improving energy efficiency.

The signal conditioning circuit was also evaluated through simulation. The band-pass filter successfully isolates signals around 20 kHz, while attenuating unwanted noise. The op-amp amplifies the low-amplitude input signal, and the peak detector accurately captures the signal envelope. The smoothing capacitor produces a stable DC output, proving that the circuit is suitable for interfacing with the microcontroller.

Overall, the simulation results validated that both circuits perform as intended, reliably processing the signals and delivering power efficiently. This ensured that the system could be implemented confidently, reducing development time and minimising hardware errors.

3.3 Software design

The software design was implemented using an Arduino to process the signal and generate control outputs for the motor drive system. The program was structured to

continuously read the analogue input from the signal conditioning circuit and convert it into a control signal for motor operation.

The core functionality of the software is to translate the detected signal strength into PWM signals that control motor speed. This allows the mouse to respond dynamically to changes in the detected magnetic field, enabling track following behaviour.

Thresholds used to determine when the signal corresponds to the track, ensuring that noise does not trigger incorrect motor responses.

The software was developed with a modular structure, separating input reading, signal processing, and motor control functions. This improves readability, debugging, and future modifications. Additionally, real-time responsiveness was prioritised to ensure that the system reacts quickly to changes in signal input, which is critical for maintaining stable movement along the track.

Testing and validation of the software were conducted alongside hardware testing to ensure correct interaction between the control algorithm and the physical system. This integration ensured that the software reliably translated sensor input into controlled motor behaviour.

4 Final Product

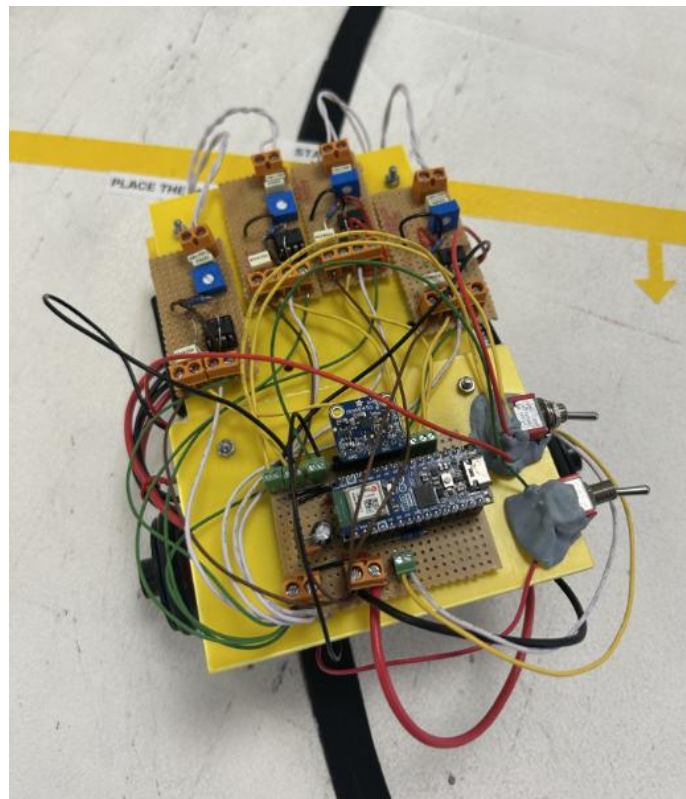


Figure 4: The final mouse system

Although the final system was unable to lap the track in a reasonable time on its own, we understood that there were significant corrections to be made, as well as optimisation strategies that we have to look into. The final system was able to follow the current-carrying track, but at turns and bends it would need repositioning, which ultimately meant that something was wrong. If we had more time and resources, we would definitely look into the issues and aim to produce a full working, successful system.

5 Investigation of complex problem with laboratory skills

During Project Week, we used laboratory equipment throughout the design, assembly, and testing stages of the mouse robot system. We used tools such as soldering equipment and electronic testing instruments to construct the circuit and verify the functionality of different components. These tools allowed our team to accurately assemble the hardware and analyse the behaviour of the electrical signals within the system. By using this equipment, we were able to develop a functioning mouse robot. Although the robot could operate and was able to move, it unfortunately did not successfully follow the black line, meaning that we had to work on the sensor control system more thoroughly.

Circuit Assembly

We made use of a soldering station during the assembly of the different electronic circuits we had for the mouse robot. We used it many times until we assembled a correctly functioning electronic board, meaning some of the boards that we soldered happened to be incorrect. The station consisted of a temperature-controlled soldering iron and the hot tip of the iron was carefully applied to both the component lead and the copper pad to ensure that there was proper solder flow and strong joints. In addition, a fan was used to remove solder fumes from the working area, improving safety and breathability during assembly. Careful soldering practices were followed throughout the process, resulting in a clean and well-assembled board with no visible solder bridges, cold joints, or other soldering defects. This ensured the circuit connections were reliable and suitable for testing of the robot system. Although there did happen to be some wasted solder on the board due to the fact that some soldering stations worked less effectively compared to others, this wasn't a necessarily major problem, and we still produced a well-soldered board for testing.

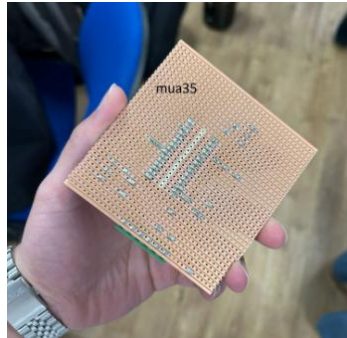


Figure 5: A soldered piece

Verifying circuits using an oscilloscope

An oscilloscope was used extensively to analyse and verify the behaviour of both the motor drive circuit and the peak detector circuit within the mouse robot system. As a team, we were aiming to achieve a lap time of 20 seconds with our mouse, and this required accurate motor control and sensing. The system was powered using a 6 V battery to drive the DC motors, and the Arduino would provide additional power to control the system how we wanted it to.

For the motor drive circuit, the oscilloscope was used to observe the control signal supplied to the motors. As shown in one of the images below, a clear square wave is visible, characterised by a frequency and duty cycle that we set using a waveform generator. This waveform represents a switching signal, used for pulse width modulation to control motor speed. The consistent amplitude and sharp transitions between high and low voltage levels confirm that the circuit is producing a stable control signal, verifying that the motor drive stage operates correctly and also ensuring the motors can be driven effectively. This also reflects the circuit that we simulated earlier on LTspice, and the fact that the simulated waveform matches the one we produced on the oscilloscope adds credibility to its accuracy.

For the peak detector circuit, the oscilloscope was used to observe both the input and output signals simultaneously. One channel displayed the sinusoidal input signal from the pick-up coil, while the second channel was used to monitor the peak detector output. The initial measurements, shown in the images, represent the system before we implemented it correctly, where the output did not consistently track the peak of the input signal. After increasing the gain of the peak detector circuit, the output signal (green trace) aligned with the peak of the sinusoidal input (yellow trace). As the sensors were moved closer to and further from the current-carrying conductor, the amplitude of the input signal varied accordingly, and the peak detector output continuously tracked these changes by maintaining a voltage level at the signal's maximum value. This behaviour confirms that the circuit was correctly detecting and holding the peak of the input signal. The ability of the output to follow changes in signal amplitude in real time verifies correct operation of the peak detector and is essential for reliable sensor signal processing in this application.

Overall, these oscilloscope measurements clearly show that both circuits are functioning correctly. By analysing key parameters such as amplitude, frequency, and duty cycle, the oscilloscope enabled real-time validation of circuit performance and ensured that the system behaviour aligned with the intended design, even though the final robot did not successfully achieve line-following.

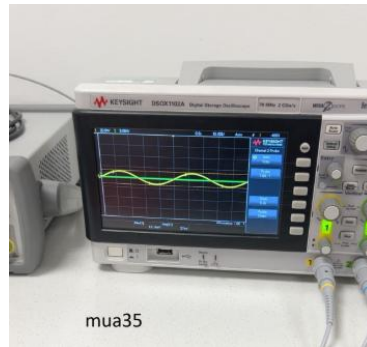


Figure 6: Peak detector input output signals

Signal generation and testing using a waveform generator

The waveform generator was used to provide controlled input signals to the circuit during testing and validation. This equipment was primarily used in the signal conditioning stage of the project, where it simulated the output of the pick-up coil and allowed the behaviour of the peak detector and motor drive circuits to be analysed under the conditions that we set. By generating repeatable signals, we had accurate observation of system response using the oscilloscope.

As shown in one of the images below, the waveform generator was configured to produce a square wave signal. This type of input was used to analyse the switching behaviour of the system, particularly in the motor drive circuit. The square wave, characterised by its frequency and 50% duty cycle, allowed the response of the circuit to rapid transitions between high and low voltage levels to be observed. This was useful for verifying that the control signals operated correctly and ensuring the system responded appropriately to digital inputs.

The second image below shows the waveform generator configured to output a sinusoidal signal. This waveform was used to simulate the real output of the pick-up coil, providing a more realistic input for testing the peak detector circuit. The smooth, continuous nature of the sine wave allowed the circuit's ability to track and extract the peak value of a varying signal to be evaluated. Observing the system under both square and sinusoidal inputs ensured that the circuit was not only functioning correctly under controlled test conditions, but also under realistic operating scenarios.

Overall, the use of the waveform generator was essential in both the development and validation stages of the project, as it enabled systematic testing of the circuit using a range of input signals relevant to the final application. We used the waveform generator in the different applications we needed it for, the peak detector and the motor drive circuits.



Figure 7: Square and sinusoidal wave generation

Using fabrication tools for robot chassis

The robot chassis required us to use fabrication tools to design a mounting component to securely position the pick-up coil sensors. To achieve this, we were given plastic boards and we shaped them using a range of workshop tools, including saws, drills with various drill bits, chisels, files, and clamps. The clamp was used to firmly secure the material during cutting and drilling, ensuring both safety and accuracy. A saw was used to cut the plastic to the required dimensions, after which holes were drilled to allow the component to be mounted onto the robot chassis using screws. Chisels and files were then used to refine and clean the drilled holes, improving the fit and finish. Any team member that was set to do this part of the project was also required to wear safety goggles before stepping into the workshop environment as standard safety practice.

This fabricated component acted as a header to hold the pick-up coil sensors in place, allowing them to be positioned as close as possible to the current-carrying wire. Maintaining this was essential for maximising signal strength and ensuring reliable sensor readings. The use of fabrication tools therefore played a crucial role in enabling accurate sensor placement and supporting the overall functionality of the robot system.

6 Sustainability

6.1 Sustainable Energy Management

Identification of energy performance improvements

Several design decisions were implemented in the project to improve the overall energy performance of the mouse robot system.

Firstly, for energy efficiency, the motor drive circuit was designed using a switch-mode control system rather than a linear control method. A MOSFET was used as a switching device and is controlled by a PWM signal generated by the Arduino microcontroller. By rapidly switching the MOSFET on and off, the duty cycle of the PWM signal controls the average voltage supplied to the motor, allowing the motor speed to be adjusted efficiently. This approach significantly improves energy efficiency because the MOSFET operates mainly in either the fully ON or fully OFF state, reducing power dissipation as heat compared with linear control methods [1].

Secondly, for energy efficiency and conservation, a band-pass filter combined with a peak detector was implemented to process the signal from the pick-up coil. The band-pass filter isolates the desired frequency of the signal from the track wire while attenuating unwanted electrical noise and interference. The peak detector then converts the filtered AC signal into a measurable DC voltage that can be read by the microcontroller. Using a peak detector improved the accuracy of the signal processing, particularly because the coil signal amplitude was very small, allowing the control system to respond more reliably to the track signal. A cleaner signal also allows the microcontroller to make more accurate steering decisions, preventing unnecessary corrections to the motor speeds. This reduces wasted motor activity and therefore lowers overall energy consumption during operation [3].

Mechanical integration improvements were considered to minimise energy losses in the system. Proper alignment of the motors, wheels, and chassis reduced friction and rolling resistance, allowing the motors to operate more efficiently. Additionally, maintaining a lightweight structure reduced the load on the motors, lowering the electrical power required to drive the robot. A lower required torque makes it easier to accelerate and climb the ramp sections of the track. Using lightweight materials such as plastic helped minimise the mechanical load on the motors, reducing energy consumption throughout the run. These mechanical considerations helped conserve energy during the robot's movement along the track. A potential further improvement could've been to replace the front caster with a low friction plastic skid which would lower the mechanical drag required to propel the mouse. This would help save energy over the 20m track.

Additionally, for energy conservation, we installed isolating switches in the battery power supply system to prevent energy drain when the mouse was idle. The mouse's differential steering system could also be improved by using a regenerative braking circuit. This would recover kinetic energy normally lost as heat when the mouse decelerates for gentle bends, corners, or when travelling down the track's falling ramp. Although clean energy isn't directly implemented onboard, the mouse relies on grid charged or disposable external power, specifically limited to single-use alkaline 9V batteries or rechargeable NiMH 1.2V batteries. To improve, we could've added miniature solar panels on the top of the chassis to power the batteries sustainably.

Overall, the combination of efficient switch-mode motor control, improved signal processing through the filter and position rectifier, and careful mechanical integration

enhanced the energy performance of the system while ensuring reliable operation of the mouse robot.

LCA diagram

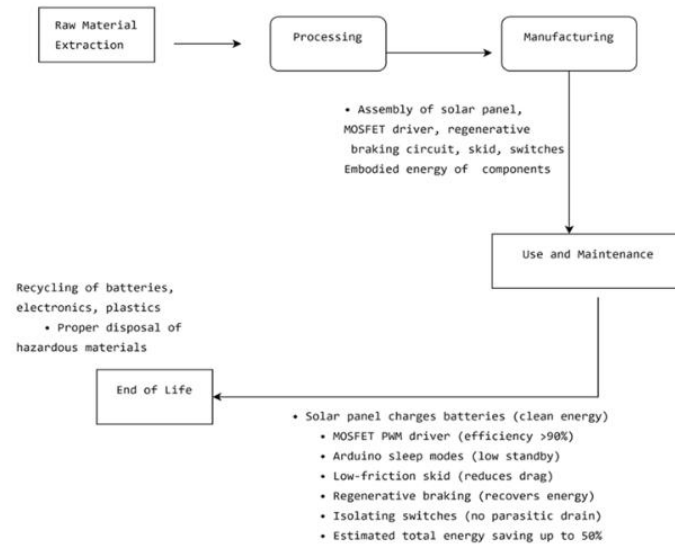


Figure 8: Mouse LCA diagram

Explanation of stages

Raw Material Extraction

Materials such as copper (for wires and coils), silicon (for semiconductors), and plastics used in electronic components were extracted from natural resources.

Material Processing

Raw materials were refined and processed into usable forms, such as semiconductor wafers, copper wiring, and plastic housings.

Manufacturing

Electronic components such as MOSFETs, resistors, capacitors, and microcontrollers were fabricated and packaged.

Assembly

The components were integrated onto the breadboard, and combined with the mechanical parts such as the motors, wheels, and the robot chassis.

Use Phase

The mouse robot operated using electrical energy from its power supply. Efficient control systems, filtering circuits, and motor drivers helped minimise energy consumption during this stage [2].

Maintenance

Components such as motors or sensors would require replacement or recalibration over time to maintain performance, however as we don't think the mouse robot would be utilised moving forward, not much maintenance would be required post project week.

End of Life

At the end of the product's lifecycle, electronic components and metals could be recycled to recover valuable materials, while non-recyclable parts such as the wires may be disposed of according to electronic waste regulations [4].

6.2 Sustainable materials management

Description

A hazardous material present in the Mouse Race project is the electrolyte within the battery types used to power the system, specifically rechargeable NiMH AA 1.2 V batteries and single-use alkaline PP3 9 V batteries. The mouse's power system needed to be portable and on-board, so the battery configuration, voltage levels, and charge state were important design considerations. These batteries were required to provide the control circuitry and motor drive system with sufficient electrical energy [9].

The electrolyte in these batteries is typically an alkaline solution [5], [6], commonly potassium hydroxide (KOH), which is used for ion transport within the cells and it enables electrochemical reactions to occur. In alkaline PP3 batteries, this is used for reactions between zinc and manganese dioxide, whereas in NiMH batteries, it enables hydrogen absorption and release within the metal hydride structure. In both cases, the electrolyte has a critical role in converting chemical energy into electrical energy. As a result, these chemically reactive electrolytes are a key hazardous material embedded within the project.

Explanation of Use

The battery electrolyte is used indirectly in the project through the batteries to provide electrical energy for all the subsystems of the mouse. This includes powering the signal conditioning circuits, control system, and DC motor drivers for movement. The mouse required sufficient voltage for speed and adequate current for torque, and this came from the battery meaning it was a critical component of the design. Without this energy source, the mouse would not be able to operate. Therefore, the electrolyte plays a vital role in enabling electrolysis to happen, the process that essentially produces usable electrical power from chemical energy for the entire system.

Explanation Hazardous or Critical

The electrolyte within the batteries is considered hazardous [8] because it contains chemically reactive substances that can be harmful if they're released into the environment. If a battery is damaged, not handled carefully, or disposed of incorrectly,

it may leak, leading to chemical exposure risks and environmental contamination [5], [8]. Batteries also contribute to electronic waste, and if they aren't disposed of correctly, it can result in toxic substances entering soil and water systems. In the context of this mouse race project, continuous testing requiring the use of batteries increases the likelihood of waste being produced, making the presence of this hazardous material significant from a sustainability perspective.

Proposed Engineering Solution

One engineering approach to improving sustainability is to reduce the environmental impact associated with battery use while also applying sustainable practices already demonstrated in the project. The mouse race project considered sustainability by reusing the chassis and mounted components from previous years, and we were advised not to excessively modify the pieces so that they could be reused and recycled again. This reduces material waste and extends the component life expectancy.

More importantly, battery-related sustainability could be improved by minimising unnecessary battery replacement through efficient circuit design and careful power management during testing. This includes optimising the system using test data to avoid excessive current being drawn, ensuring components operate within appropriate voltage ranges, and limiting long and inefficient testing cycles that would cause the battery to deplete faster. In addition, a key practical approach that we can take is physically protecting the mouse system during testing. If we are careful with the mouse during test rounds, we can avoid damaging it if it goes off the track, and in situations where it deviates off course, we should promptly intervene to ensure that it doesn't fall off the table or collide with obstacles such as walls. Being vigilant is particularly important in protecting the batteries, as physical damage could lead to the potassium hydroxide (KOH) electrolyte leaking, posing both safety and environmental risks.

By combining careful operational practices, efficient energy usage, and the reuse of existing hardware, the overall environmental impact associated with hazardous battery materials can be significantly reduced while maintaining a reliable system performance.

Explanation of why material is necessary

The properties of the battery electrolyte are effectively irreplaceable within this project because it has a unique role in electrolysis. The electrolyte, typically potassium hydroxide (KOH) in alkaline batteries, provides a medium for ion transport between the zinc (Zn) anode and manganese dioxide (MnO_2) cathode, allowing the reactions that generate electrical energy to occur. Without this ionic conduction, electrons can't flow through the external circuit, and the battery cannot produce power. Alternative materials can't replace the electrolyte without actually changing the battery chemistry or system design. Therefore, due to its high ionic conductivity, chemical stability, and compatibility with electrode materials makes it essential and irreplaceable within the design.

Significant potential impact on sustainability

The selected hazardous material, the electrolyte within the AA and PP3 batteries, is fully embedded within the mouse race project and directly influences the system's overall performance and sustainability. The batteries power all the subsystems, including the control circuitry and motor drive, meaning their behaviour affects speed, torque, efficiency, and reliability. As we repeat the tests, battery usage increases, leading to higher material consumption and potential environmental impact through waste and disposal. Therefore, the electrolyte is not a peripheral material but a core component of the design. Improving its management through efficient power usage and reduced amount of replacement has a major impact on the overall sustainability of the project.

6.3 Engineering design for circular economy

The R investigated and technology used

In the project, we applied circular economy principles by considering the 4Rs [13] (Reduce, Reuse, Recycle and Repair), and we considered them across both the electrical and mechanical aspects of the design.

Reduce was implemented in several ways. Electrically, we used an efficient MOSFET-based PWM motor driver instead of a linear control circuit, and this significantly reduced energy losses by operating in a switching manner (on and off) rather than dissipating power as heat. Signal processing was optimised using a pick-up coil with a peak detector circuit, which minimised unnecessary computational and hardware complexities. Mechanically, we reduced material usage by avoiding a full protective plastic casing outside the mouse, but instead made sure to be careful with it during testing, which reduced plastic usage without compromising the mouse's functionality. Other ways in which reduce was implemented was through design features such as low-friction skids and efficient control strategies, which lowered overall energy demand.

Reuse was embedded in the design by selecting modular and recoverable components. The Arduino Nano ESP32, DC motors, and sensor circuits can be disassembled and reused in other projects. Furthermore, we reused a plastic header from last year and incorporated it into the mouse chassis, showing how we reused components mechanically as well. This component was attached using screws rather than permanent bonds, and this allows it to be easily removed and reused by the cohort next year, extending its lifecycle and reducing demand for extra materials. Also, instead of soldering the DC motor connections directly onto the stripboard, we used pin headers (connector interfaces) and soldered them onto the board. This allowed the motor circuits to be connected and disconnected without a permanent connection, enabling the motors and associated wiring to be reused without damaging the main circuit.

Recycle was considered at the end-of-life stage through material recovery pathways. The circuit was implemented on a stripboard (copper prototyping board), which contains continuous copper tracks or conducting lines. While soldered components are not easily reusable, the board itself contains copper which is a valuable metal that can be recycled through electronic waste processing. Additionally, wires, connectors, and plastic elements can be separated and processed according to WEEE (Waste Electrical and Electronic Equipment) guidelines, ensuring that materials are responsibly disposed and recovered. This will reduce landfill waste, and it recovers embedded resources from the system.

Repair was also considered through the design's accessibility and modularity. Components such as motors, sensors, and circuit elements were arranged in a way that facilitates individual parts to be changed without requiring the system to be completely disassembled and redesigned. This extends the functional lifespan of the system and reduces the need for full replacement so less components are wasted. Overall, by integrating these principles, we were able to reduce environmental impact while maintaining a good system performance and functionality.

Impact on the LCA (CO2 footprint)

The life cycle assessment of the mouse considers six key stages. Each stage contributes differently to the overall carbon footprint, and I have shown them below.

Raw Material Extraction and Processing

The initial stage involves extracting and processing raw materials such as copper (used in the wiring, coils, and stripboard tracks), silicon (for the semiconductors within the Arduino Nano ESP32 and MOSFETs), and plastics (for insulation and chassis components). These processes require a lot of energy due to mining, purification, and semiconductor fabrication, all of which need significant thermal and electrical energy. From what I have found on documented sources, this stage contributes approximately 0.2 - 0.3 kg CO₂ to the overall footprint for small-scale electronic systems like this mouse. This shows how important material reduction is, as reducing the amount of materials used directly lowers CO₂ emissions at this stage.

Manufacturing and Assembly

The manufacturing stage involves producing the electronic components such as the Arduino, the MOSFETs, passive components, and DC motors, as well as assembling these components onto a stripboard (copper prototyping board) and then integrating into the mouse chassis. The fabrication of semiconductors and electronic devices is especially carbon intensive due to cleanroom processing and high-energy manufacturing techniques. Additionally, the soldering and assembly processes create fumes which contribute to more emissions. This stage is estimated to contribute approximately 0.4 - 0.6 kg CO₂, and it is the largest contributor to the carbon footprint. Small electronic devices contribute significant embodied carbon due to manufacturing processes [11]. This reinforces the importance of reuse, as avoiding repeated manufacturing significantly reduces the overall emissions.

Distribution

The distribution phase is where the transportation of components from the manufacturers to the users happens, including packaging and logistics. Although the individual components are lightweight, the main emissions arise from fuels during transport and materials that are used in packaging. As the main system is relatively small, this stage contributes a lower amount of approximately 0.05 kg CO₂. While this number is smaller compared to the emissions in other stages, emissions in this stage can be reduced through efficient supply chains and bulk transportation.

Usage Phase

During operation, the mouse consumes electrical energy supplied by the battery system [15]. For this stage we have to have an estimate for the runtime of our system rather than assuming that the mouse will be used for years to come, as this will give an accurate figure for the emissions at this stage. As a team, we were aiming for a lap time of 20 seconds, and from what we recorded, our mouse completed one full lap (approximately 20 seconds considering that we had to continuously put it back on its track) and two additional half-lap test runs (approximately 10 seconds each), resulting in a total active runtime of approximately 40 seconds (0.011 hours).

Our operating values were:

- Voltage = 6 V
- Current approximately 0.6 A
- Power = 3.6 W

The total energy consumption is:

$$E = 3.6 \times 0.011 = 0.04 \text{ Wh} = 0.00004 \text{ kWh}$$

Using the UK emission factor of 0.233 kg CO₂ / kWh [10], the resulting emissions are:

$$\text{CO}_2 = 0.00004 \times 0.233 = 0.0000093 \text{ kg CO}_2$$

This value is very small, which shows that the carbon footprint during the mouse's operation is effectively negligible. Even when accounting for additional minor tests or idle periods or even incorrect runtime estimates, the contribution from the usage phase will be insignificant compared to the emissions from other stages like material extraction and manufacturing.

This reinforces a key conclusion of the life cycle assessment, which is for small-scale electronic systems, the main environmental impact comes from production rather than operation, and this again shows why reduce and reuse strategies are important in the design.

Energy Efficiency Improvements (Design Impact)

The design includes many features that significantly reduce energy consumption during operation. The use of PWM motor control (which has greater than 90% efficiency)

minimises energy losses compared to linear control [12], which can dissipate up to 30 - 50% of energy as heat. Additional features such as low-friction skids (reducing mechanical resistance), Arduino sleep modes (minimising standby power), regenerative braking (recovering a bit of kinetic energy), and isolation switches (eliminating small current draws) further enhance efficiency. All of these together reduce the overall energy consumption by up to 50%, effectively halving the usage-phase emissions to approximately 0.00000465 kg CO₂. While the absolute reduction is small, it shows how optimisation strategies can help with sustainability.

End-of-Life and Reuse

At the end-of-life stage, the environmental impact depends on the future of the mouse. The design prioritises reuse, with key components such as the Arduino Nano ESP32, motors, connectors and plastic headers being reusable in future projects. This avoids additional manufacturing emissions, which would otherwise contribute approximately 0.4 - 0.6 kg CO₂ per reuse cycle. For non-reusable elements, such as the soldered components on the stripboard, there are recycling pathways. The copper tracks within the stripboard can be recovered through electronic waste processing, while plastics and wiring can be separated and processed under WEEE regulations. If the components aren't disposed of correctly, this stage may contribute an additional 0.1 - 0.2 kg CO₂ [11], but effective reuse and recycling methods significantly mitigate this impact.

Total CO₂ Footprint

Combining the estimated emissions from all the lifecycle stages, the total estimated carbon footprint of the mouse is approximately 0.7 - 1.0 kg CO₂. The analysis above clearly shows that manufacturing and material extraction dominate the emissions, while the usage phase contributes an extremely small amount. This highlights the importance of circular economy strategies such as Reduce and Reuse, as minimising material use and maximising component lifespans have the greatest impact on reducing the overall emissions.

Monetary value created

The circular design approach in this project generates economic value by reducing both initial and lifecycle costs. The total cost of a typical system that uses fresh components is estimated to be approximately £40 - £70 [14], including an Arduino Nano ESP32 (£15 - £25), DC motors (£10 - £20), batteries (£5 - £10), and additional electronic components (that would sum up to about £10 - £15) [14]. By applying circular economy principles, we saved significant costs. Reusing components such as the microcontroller, motors, and connectors avoids repurchasing costs, while efficient circuit and mechanical designs reduces energy consumption and material usage costs. Overall, I estimated the lifecycle cost saving to be in the range of £30 - £60 for each mouse, which represents a reduction of up to 50 - 70% compared to a fully new system. This shows that circular design also has serious economic benefits as well as environmental impacts.

Monetary value from Reduce

The Reduce principle generates monetary value by minimising both material usage and energy consumption. By implementing a PWM motor control system, we improved efficiency by approximately 30 - 50% compared to linear control circuits, reducing wasted energy and extending battery life. This reduces how often we have to change batteries, resulting in estimated savings of £5 - £10 over repeated use cycles. Additionally, we reduced material usage by avoiding a full protective casing which removed the need for extra plastic components, saving approximately £3 - £8 per build. Efficient circuit design, including optimised signal processing using the peak detector, also reduces the number of components and wires required, reducing the overall material costs by approximately £5 - £10. These reductions collectively contribute to a lower total system cost and they improve resource efficiency.

Monetary value from Reuse

Reuse creates the largest economic benefit by extending the lifespan of key components and removing the need for repurchasing them. Reusing the Arduino (£15 - £25), DC motors (£10 - £20), and other electronic components significantly reduces costs for future projects in university. By including pin headers (connector interfaces) instead of directly soldering wires allows the motors and subsystem circuits to be disconnected and reused without damage, therefore preserving their value. Furthermore, the use of the reusable plastic header, attached using screws, ensures that it can be recovered and reused by the next-year cohort, saving additional material costs. Overall, reuse contributes an estimated £25 - £45 in avoided costs per reuse cycle, and this makes it the most economically impactful circular strategy in this mouse project design.

Monetary value from Recycle

The Recycle principle creates economic value by recovering materials at the end-of-life stage. Although some components are soldered and not directly reusable, the stripboard (copper prototyping board) contains recoverable copper, which has market value as a recyclable material. Copper scrap prices typically range from £4 - £6 per kg, and while the quantity in this system is quite small, it still has a recoverable value rather than waste. Additionally, recycling of plastics and wiring reduces disposal and electronic waste management costs. While the direct financial return isn't much (estimated less than £1 per system), recycling reduces costs that would otherwise go towards material extraction and processing, contributing indirectly to economic sustainability.

Monetary value from Repair

The Repair principle creates value by reducing the need for replacing the system completely when there has been an error. The modular design through the use of discrete components and connector interfaces, makes it possible for faulty components such as motors or sensors to be replaced at lower cost rather than having to rebuild the entire system from scratch. For example, replacing a single DC motor (approximately £5 - £10) is significantly cheaper than replacing the full system (approximately £40 - £70). This reduces maintenance costs and extends the usable life of the system. Over multiple use cycles, repairability can result in an estimated savings of approximately £10 - £20, depending on component wear and failure rates.

7 Risk and change management

7.1 Risk mitigation

Severity (a)		Likelihood of Occurrence (b)	
1 – Trivial completion	Slight deviation in line following, no impact on lap	1 – Remote	<10% chance of occurring during the week
2 – Minor	Mouse leaves track but recovers, slight time loss	2 – Unlikely	11-30% chance of happening during the week
3 – Moderate	Unable to set a competitive time	3 – Possible	31-50% chance of happening during the week
4 – Serious	Failure to complete laps consistently	4 – Likely	51-80% chance of happening during the week
5 – Catastrophic	Non-performing mouse or destruction of components.	5 – Very likely	>80% chance of happening during the week

Risk Assessment Matrix					
(B)↓ (A)→	Trivial	Minor	Moderate	Serious	Catastrophic
Remote	1	2	3	4	5
Unlikely	2	4	6	8	10
Possible	3	6	9	12	15
Likely	4	8	12	16	20
Very likely	5	10	15	20	25

Risk Rating Bands (A x B)		
LOW RISK	MEDIUM RISK	HIGH RISK
Continue, but review periodically to ensure controls remain effective	Continue, but implement additional reasonably practicable controls where possible and monitor regularly	STOP THE ACTIVITY Identify new controls. Activity must not proceed until risks are reduced to a low or medium level

#	Hazard(s) identified	Existing controls & measures	Severity (a)	Likelihood (b)	Risk Rating (a x b)	Additional control/action required	Severity (a)	Likelihood (b)	Risk Rating (a x b)
1	Integration faults: Modular circuits do not interface correctly on Friday morning.	Communicate what we are working on and decide what approach we are taking between analogue and digital early on.	4	4	16	Use the "modular construction" approach and establish a clear functional block diagram within the first hour.	4	2	8
2	Momentum overload: Mouse gathers too much speed and shoots off the final corner.	Work out the minimum torque required to get over the hill, so there won't be enough momentum to shoot off	5	3	15	Implement real-time position feedback control to slow the "outer" motor or reduce overall PWM duty cycle during curves.	5	1	5
3	Sensor noise: 20kHz signal is drowned out by ambient electronic noise, causing loss of track.	Implement the passive resonant bandpass filter and peak detector as specified in the brief.	3	4	12	Conduct extensive tests and keep the bandpass pass wide enough so that it doesn't malfunction when using slightly different signal strengths.	2	3	6
4	Inadequate drive power: Mouse lacks torque to ascend the final rising ramp.	Work out the maximum torque the motor can produce considering the weight of our mouse	4	3	12	Test wheel speed under load using the optical tachometer and ensure 6V battery supply is fully charged.	4	2	8
5	Component failure: Power MOSFET or Op-amp burnout during testing due to wiring errors.	Use reasonable power supplies.	4	3	12	Double-check wiring with a teammate and use current-limited lab power supplies for initial tests.	4	1	4
6	Time management: Not completing mouse for Friday due to poor communications and teamwork	Set deadlines for tasks to be completed and prioritising between written tasks and technical ones.	5	2	10	Establish clear protocols and team management techniques to allow clear communication of what is being worked on and what needs doing.	5	1	5
7	Trivial misalignment: Slight sensor tilt causing a minor 0.2s delay in lap time.	Visual check of the sensors before use	2	2	4	Measure the angle of the sensors prior to use to keep a consistent alignment.	1	1	2

Figure 9: Risk assessment matrix

7.2 Change management

GitHub repository

GitHub repository: <https://github.bath.ac.uk/mua35/PW3-Mouse-Robot>

I created a GitHub repository to manage and track the development of the mouse control system. The repository contains the Arduino code that was used for the motor control and navigation. It serves as a central platform for version control, allowing changes to be tracked, documented, and reverted if necessary.

The commit history above demonstrates how changes were tracked throughout the development process. I added a descriptive message with each commit, allowing clear documentation of updates such as the initial setup, file uploads, and the transition from a dual-sensor to a single-sensor control system.

Managing a change

A significant change was made to the Arduino code and the control logic, transitioning from a dual-sensor system to a single-sensor approach. This was due to the fact that for some reason which we still haven't figure out, only one of the DC motors were running on race day when previously on the day before, both motors and circuits were operating perfectly. Therefore, this change was needed so that we had a working mouse robot to present for the race. The code required modifications to PWM parameters, motor speed limits, and control behaviour. We had to test the DC motors using the electronic measurement equipment mentioned before to find these parameters. The changes are clearly shown in the commit comparison, where removed code (red) and newly implemented logic (green) demonstrate how the system was updated and refined. This screenshot is only part of the code, but a lot more has changed beyond this.

Branch and merge workflow

To accurately implement the change, I created a new branch named `single-sensor-mouse-update` from the main branch. This allowed the development and testing of the updated control logic without affecting the stable version of the code. This was good because if I made any mistake, the original copy was still safe.

Once the changes were completed, a pull request was created to merge the updated branch back into the main branch. This workflow ensured that all modifications were reviewed and properly integrated, and this helped maintain version control and reduce the risk of introducing errors into the main system.

Documenting the change

The change was documented using commit messages and pull request descriptions. The update included a clear explanation of the technical changes, such as changes to sensor configuration, calibration methods, and PWM control strategy. This

documentation strategy ensures that if or when I come back to this in the future, I can understand what was changed, why it was changed, and how it affects system performance.

8 Feasibility Study

8.1 Design and Development matrix

A D&D matrix was created to structure the design process and ensure that user needs were systematically addressed into target specifications and corresponding solutions. The matrix is provided in Appendix A.

The D&D matrix establishes clear links between some identified user requirements, target specifications, and technical solutions. Each requirement had measurable objectives with defined and realistic tolerances, allowing us to evaluate performance during testing. This ensured that design decisions were not arbitrary, but instead directly aligned with the functional goals of the mouse, such as reliable signal detection, controlled motor performance, and stable mechanical integration.

Many alternative solutions were considered for each requirement before selecting the most appropriate approach based on feasibility, performance, and project constraints. For example, different methods of motor control, signal detection, and chassis construction were evaluated through decision-making matrices to balance efficiency, reliability, and simplicity. This comparative approach supported the team with informed decision-making, and it reduced the likelihood of design flaws during implementation.

The matrix also helped with project organisation by clearly defining design priorities and providing a reference for tracking progress throughout the development process. By linking requirements to testing criteria, it enabled us to effectively verify if the final system met the correct requirements.

Overall, the D&D matrix ensured a structured, traceable, and systematic design approach during the week, improving both the quality of the final system and the efficiency of the development process.

8.2 Decision making matrices (including description of criteria)

We developed a set of structured decision-making matrices to explain our design decisions within the mouse robot project. These matrices are presented in Appendix B. The purpose of these matrices were to help us objectively compare between multiple design options by evaluating them against consistent criteria such as performance, efficiency, feasibility, reliability, and ease of implementation.

For each subsystem, such as motor control, signal detection, and mechanical design, there were several potential solutions that we assessed. Each option was scored

against a set of criteria criteria, allowing us to clearly visualise trade-offs between factors. For example, in the motor control design, switching (PWM-based) control methods were compared against linear control, where PWM was selected due to its significantly higher energy efficiency and lower power dissipation. Similarly, different signal conditioning approaches were evaluated, with the band-pass filter and peak detector combination chosen for its ability to reliably isolate the 20 kHz track signal while minimising noise.

The use of decision matrices ensured that design choices were not based solely on intuition and resources that were provided to us, but instead on a transparent and repeatable evaluation process. It also allowed the team to justify decisions with clear engineering reasoning, particularly when balancing conflicting requirements such as simplicity against performance or efficiency against complexity.

Furthermore, the matrices improved team-collaboration by providing a common framework for engagement and decision-making. By quantifying each option, we were able to resolve disagreements using evidence rather than subjective preference. This aligns with the project's structured approach to engineering design and contributes to a more traceable development process.

Overall, the decision-making matrices in Appendix B played a critical role in helping team members understand and justify why certain solutions were chosen, ensuring that the selected options were optimal across the many criteria and aligned with the project's performance and sustainability objectives.

9 Conclusion

This project demonstrated the design, development, and testing of an autonomous mouse robot capable of detecting a 20 kHz track signal and converting it into controlled motor movement. The integration of signal conditioning, microcontroller-based control, and a PWM-driven motor system resulted in a functional prototype that exhibited stable behaviour during testing. Simulation tools such as LTspice played a key role in validating circuit performance prior to implementation, reducing design errors and improving overall system reliability.

Throughout the project, a range of technical skills were developed, including circuit design, simulation, signal processing, and embedded programming. Practical laboratory skills were strengthened through the use of oscilloscopes and waveform generators to verify system behaviour. In addition, we gained experience in system integration, where electrical, mechanical, and software components were combined to achieve a working solution. Teamworking and communication were also critical, particularly in coordinating tasks, resolving design challenges, and maintaining progress within time constraints.

Overall, the project contributed significantly to both our development by reinforcing a structured engineering design approach. It highlighted the importance of iterative testing, problem-solving, and efficient design practices when working on complex systems. The experience gained provides a strong foundation for future engineering projects, particularly in areas involving control systems, electronics, and multidisciplinary design.

References (including Quality Referencing)

[1] A. G. Olabi et al., “Battery electric vehicles: Progress, power electronic converters, strengths, weaknesses, opportunities and threats,” *ScienceDirect*, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666202722000751>. [Accessed: 22 Apr. 2026].

Strong: This peer-reviewed journal gives high-quality and up-to-date analysis of power electronics and energy efficiency in electric systems. It is directly relevant to PWM motor control and energy optimisation, discussing efficient switching methods that we could use in the project.

[2] N. Khan et al., “A critical review of battery cell balancing techniques,” *ScienceDirect*, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2352484724002506>. [Accessed: 22 Apr. 2026].

Strong: This recent review paper provides knowledge about battery performance, efficiency, and energy management strategies. It is relevant for discussing energy use, efficiency, and optimisation in battery-powered systems.

[3] Z. E. Eđer, “Reinforcement learning-based energy management strategy for fuel cell hybrid electric vehicles,” MSc thesis, 2022. [Online]. Available: https://research.sabanciuniv.edu/id/eprint/42949/1/10439555_Eger_Zekeriya_Ender.pdf. [Accessed: 22 Apr. 2026].

Strong: This academic report provides a good discussion about energy management strategies and efficiency optimisation in electrical systems. It supports reducing energy consumption through control strategies, which supports our use of PWM and efficient system design.

[4] “Towards a Smarter Battery Management System,” MDPI, 2023. [Online]. Available: https://mdpires.com/bookfiles/book/11066/Towards_a_Smarter_Battery_Management_System.pdf. [Accessed: 22 Apr. 2026].

Moderate: This source provides a useful background on battery systems and their role in sustainable energy management. It is informative and relevant, but it is broader in scope and less directly focused on small-scale systems like the mouse.

[5] “Alkaline battery,” *Wikipedia*. [Online]. Available: https://en.wikipedia.org/wiki/Alkaline_battery. [Accessed: 6 Apr. 2026].

Moderate: This source provides a clear overview of alkaline battery composition and operation, including information about electrolytes. While it is useful as a base for understanding, it is not peer-reviewed which reduces its reliability although it is broadly accurate and up-to-date.

[6] “Alkaline battery chemistry FAQ,” *PowerStream*. [Online]. Available: <https://www.powerstream.com/BatteryFAQ-alkaline.html>. [Accessed: 6 Apr. 2026].

Moderate: This industry source gives detailed explanations of battery chemistry and

practical behaviour which is good for electrolyte understanding. However, it lacks academic peer review and formal referencing, which limits its credibility.

[7] N. Wang *et al.*, “A review of zinc-based battery from alkaline to acid,” *ScienceDirect*, 2021. [Online].

Available: <https://www.sciencedirect.com/science/article/pii/S2590049821000199>.

[Accessed: 6 Apr. 2026].

Strong: This peer-reviewed journal article provides analysis of battery chemistry and electrochemical processes. It is highly relevant and academically reliable, making it a strong source for electrolyte function explanations.

[8] “Battery recycling and disposal,” *Wikipedia*. [Online].

Available: https://en.wikipedia.org/wiki/Alkaline_battery#Recycling_and_disposal.

[Accessed: 6 Apr. 2026].

Moderate: This source gives an overview of battery disposal and environmental impact which is good for sustainability considerations. However, it is not peer-reviewed, so it is not the most credible.

[9] International Energy Agency, “Batteries and energy storage,” [Online].

Available: <https://www.iea.org>. [Accessed: 22 Apr. 2026].

Strong: This source provides authoritative knowledge about battery technologies and energy systems. It is highly reliable and relevant for supporting discussions about battery performance, sustainability, and lifecycle impact.

[10] UK Government, “Greenhouse gas reporting: conversion factors 2024,” [Online].

Available: <https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2024>. [Accessed: Apr. 15, 2026].

Strong: This official government source provides accurate and up-to-date emission conversion factors. It is highly reliable and directly supports quantitative CO₂ analysis, which is good for the LCA and environmental impact calculations.

[11] E. Özkan, N. Elginoz, and F. Germirli Babuna, “Life cycle assessment of a printed circuit board manufacturing plant,” *Environmental Science and Pollution Research*, vol. 25, no. 27, pp. 26801–26808, 2018. [Online]. Available: <https://doi.org/10.1007/s11356-017-0280-z>. [Accessed: Apr. 15, 2026].

Strong: This peer-reviewed journal provides detailed LCA data for electronic components. It is highly relevant and academically strong, making it a good source for evaluating environmental PCB impacts in the project.

[12] International Energy Agency, “Energy efficiency of electric motor systems,” [Online]. Available: <https://www.iea.org>. [Accessed: Apr. 15, 2026].

Strong: The IEA is a globally recognised authority on energy systems. This source provides credible data on motor efficiency, directly supporting our understanding of PWM motor control and energy optimisation strategies in the project.

[13] European Commission, “Circular economy action plan,” [Online].

Available: <https://ec.europa.eu/environment/circular-economy/>. [Accessed: Apr. 15,

2026].

Strong: This source provides a detailed framework for circular economy principles. It is highly credible and relevant for design strategies such as reuse and recycling within the project.

[14] RS Components, “Arduino Nano ESP32 and electronic component pricing,” [Online]. Available: <https://uk.rs-online.com>. [Accessed: Apr. 15, 2026].

Moderate: This industry source provides accurate pricing data for the components used in the project. While it is reliable for estimating costs, it lacks academic credibility and is less suitable for theoretical environmental analysis.

[15] International Energy Agency, “Batteries and energy storage,” [Online]. Available: <https://www.iea.org>. [Accessed: Apr. 15, 2026].

Strong: This source provides authoritative insight into battery technologies and energy storage systems. It is highly relevant for evaluating the environmental and performance impacts of batteries within the mouse’s lifecycle.

[16] STMicroelectronics, “STP55NF06 Power MOSFET Datasheet,” [Online]. Available: <https://www.st.com/resource/en/datasheet/stp55nf06.pdf>. [Accessed: 22 Apr. 2026].

Strong: This manufacturer datasheet provides detailed electrical information and characteristics of a power MOSFET. It is not the exact same as the IRFZ44N, but it is similar enough to support analysis of switching behaviour, conduction losses, and PWM-based motor control. It is highly reliable and directly supports our analysis of PWM switching efficiency and power losses in the motor drive circuit.

Appendices

Appendix A

Needs analysis		Target specifications			Technical solution		Verification		
Ref Need	Description	Ref Spec	Description	Objectives measurable + tolerance	Possible solutions	The chosen one	Verification test description	Location/Team	Control methodology
N1	PERFORMANCE - Reduce lap time by minimising friction between the robot and track surface during motion.	SP1	Lap time performance	Complete lap in 20s +/- 5s	4-wheel chassis (higher stability, higher friction) 2-wheel + skid design - Ball caster wheel design	2-wheel drive with rear low-friction skid. Selected based on speed (40%), friction reduction (30%), stability (20%), simplicity (10%). Reduces contact area so directly improves lap time target	Time full lap using stopwatch over 5 trials and calculate average lap time	Group 18	Use fully charged battery for all runs, same track layout, identical start position, and no external interference
N2	PERFORMANCE - Accurately detect the electromagnetic signal from the track wire without distortion.	SP2	Signal detection accuracy	Detect signal with > 90% reliability across during testing using testing tracks	- Simple diode rectifier - Precision rectifier circuit - Direct AC signal processing	Precision rectifier + peak detector. Chosen based on accuracy (45%), signal clarity (25%), noise reduction (20%), complexity (10%). Meets detection reliability requirement better than simple rectifier.	Compare detected signal vs known track signal using oscilloscope and count correct detections	Group 18	Reliability percentage calculation (>90%), multiple track positions tested
N3	PERFORMANCE - Deliver sufficient power to motors without excessive energy loss during operation.	SP3	Motor efficiency and control	Maintain speed with < 10% drop over 5 runs	- Linear transistor motor driver - MOSFET switching driver - Direct battery connection	MOSFET switching motor driver. Chosen based on efficiency (40%), thermal performance (25%), speed control (20%), implementation (15%). Minimises power loss vs linear driver.	Measure speed consistency over 5 runs using tachometer or lap timing	Group 18	% speed drop calculation (<10%), controlled battery conditions

N4	PERFORMANCE - Achieve smooth and controllable motor speed to prevent sudden acceleration and loss of track.	SP4	Motor control smoothness	Use tachometer to analyse rpm. Maintain stable motion with no sudden speed spikes and < 10% variation in speed during turns.	- Fixed voltage control - PWM (pulse width modulation) control - Manual resistance-based speed control	PWM motor control. Chosen based on control precision (40%), efficiency (25%), responsiveness (20%), implementation complexity (15%). Provides smooth speed adjustment, directly improving control stability and track following.	Observe motor response to speed changes and turning behaviour during track run	Group 18	Qualitative smoothness + RPM variation measurement (<10%)
N5	USABILITY - Allow quick repositioning of the sensing coil during testing to optimise wire detection.	SP5	Ease of adjustment	Adjustment completed in < 2 min without full rebuild	- Permanently soldered coil to PCB - Mechanical mount (screws/bracket) - Temporary fixing using sellotape + crocodile clips	Temporary mounting using sellotape and crocodile clip connections to the main circuit. Chosen based on speed of adjustment (40%), flexibility (30%), no permanent modification (20%), simplicity (10%). Enables quick repositioning during iterative testing.	Time how long it takes to reposition sensing coil during testing	Group 18	Stopwatch measurement (<2 min), repeat across multiple adjustments
N6	BUSINESS - Minimise cost by selecting affordable and readily available components.	SP6	Cost efficiency	Total cost of the system to department < £30 +/- £5	- Custom PCB design - Off-the-shelf components - High-performance specialised parts	Off-the-shelf lab components. Chosen based on cost (40%), availability (30%), performance (20%), ease of integration (10%). Meets cost constraint.	Calculate total cost using BOM, and receipts from online sources	Group 18	Compare total cost against £30 +/- £5 constraint
N7	HEALTH AND SAFETY - Prevent fumes from being inhaled during soldering of components onto stripboard.	SP7	Safe assembly practice	Maintain safe working distance > 30 cm from solder point and ensure visible fume displacement	- Wear respirator mask - Use extraction fan - Maintain safe distance from solder point	Use of local fan to direct fumes away from user. Chosen based on effectiveness (40%), practicality (30%), availability (20%), comfort (10%). Reduces inhalation risk during build.	Observe soldering process and confirm fumes are directed away from user in the direction of the fan	Group 18	Visual inspection + adherence to safe working distance (>30 cm)

N8	SUSTAINABILITY - Reduce waste by reusing electrical components across multiple design iterations instead of discarding parts.	SP8	Material sustainability	> 80% components reused across builds	• Single-use soldered design • Fully modular reusable design • Hybrid reuse approach	Hybrid reuse approach (reuse key components, rebuild minor sections). Chosen based on sustainability (40%), practicality (30%), time efficiency (20%), performance retention (10%).	Record number of reused vs new components across builds	Group 18	Percentage calculation (>80% reused components)
N9	AESTHETIC - Maintain a neat and organised wiring layout to improve readability of the connections and for professionalism.	SP9	Visual clarity	No loose wires > 3cm, clear routing visible	• Loose wiring • Cable-tied routing • Fully enclosed casing • No excessively long cut wires	Reduce and remove unnecessary wire length. Chosen based on clarity (35%), ease of debugging (30%), aesthetic quality (20%), accessibility (15%).	Visually inspect wiring layout and measure loose wire lengths using ruler. Assess clarity of routing and accessibility for debugging.	Group 18	Use same circuit setup and component layout across tests, assess under consistent lighting conditions, and apply identical measurement method for wire length.
N10	SOCIETAL / INCLUSION - Ensure all team members can understand and operate the robot regardless of technical background.	SP10	Accessibility of operation	All members operate system after < 10 min explanation	• Complex undocumented system • Basic labelled system • Fully documented + labelled system on GitHub and Teams	Fully documented and labelled system. Chosen based on clarity (40%), ease of use (30%), team inclusivity (20%), time efficiency (10%).	Allow new user to operate robot after explanation and observe success	Group 18	Time-based evaluation (<10 min learning), task completion check

N11	STANDARDS - Ensure the electrical system operates within safe limits aligned with recognised standards.	SP11	Electrical safety compliance	Operate at < 6 V DC, no exposed conductors	• No compliance consideration with standards • Basic safe design • Standards-informed design	Designed in line with BS EN 61010-1 (lab electrical safety), with consideration of RoHS Directive 2011/65/EU and WEEE Directive 2012/19/EU. Chosen based on safety (40%), compliance (30%), professional practice (20%), documentation (10%).	Inspect circuit for exposed conductors and measure operating voltage	Group 18	Voltage verification (<6V DC) + visual safety inspection
N12	INCLUSION / DIVERSITY - Ensure all team members can contribute ideas during design discussions without exclusion or dominance by a single member.	SP12	Inclusive communication and teamwork	Each team member contributes at least 1 idea per meeting, with no single member contributing > 50% of total discussion input	• Unstructured discussion (free conversation) • Rotating speaking turns system • Designated team leader controlling discussion	Rotating speaking turns during discussions. Chosen based on inclusivity (40%), fair participation (30%), idea diversity (20%), ease of implementation (10%). Ensures balanced contribution and aligns with inclusive teamwork objectives.	Record team meeting contributions from each member	Group 18	Contribution tracking (>1 idea per member, <50% dominance). Project Management lead in charge of keeping track.
N13	ENVIRONMENTAL - Reduce plastic waste when fabricating the low-friction skid.	SP13	Material sustainability	Ensure > 80% of skid material is reused or repurposed, with minimal new plastic used (< 20%)	• Use a new acrylic plastic sheet • Reuse skids from previous year projects • Use scrap/offcut plastic from workshop stock	Use scrap/offcut plastic from workshop stock. Chosen based on waste reduction (45%), material availability (25%), environmental impact (20%), cost (10%). Minimises new material usage while maintaining adequate performance.	Track source of skid material (new vs reused). From a workpiece, calculate percentage used via weight of skid/weight of all material.	Group 18	Percentage reused (>80%), estimate new material usage (<20%)

N14	HEALTH AND SAFETY - Safely manufacture the low-friction skid using appropriate tools and methods in the workshop.	SP14	Safe fabrication practice	Complete fabrication with 0 injuries/incidents and maintain controlled cutting (no uncontrolled fractures or slips)	• Use clamp and saws for controlled cutting • Bend plastic and snap along scored line • Use laser cutter	Use clamp and saws for controlled cutting. Chosen based on safety (40%), control/precision (30%), availability (20%), material suitability (10%). Provides stable, controlled cutting and reduces risk of sudden breakage compared to snapping, while being more accessible than a laser cutter.	Monitor fabrication process and record any incidents	Group 18	Zero incident requirement + controlled cutting verification. Adherence to workshop etiquette and rules.
N15	PERFORMANCE, USABILITY, SUSTAINABILITY - Reduce overall mouse weight to improve energy efficiency and ease of handling during testing.	SP15	Energy efficiency and ease of use	Total robot mass < 500 g +/- 50 g, with consistent performance (< 10% speed drop over 5 runs)	• Compact and tightly integrated design • Use lighter skid material • Wirelessly power the system	Compact and tightly integrated design. Chosen based on weight reduction (40%), practicality (30%), reliability (20%), implementation feasibility (10%). Reduces unnecessary mass without introducing complexity or instability from wireless power.	Measure total mouse mass using scale and test performance over 5 runs	Group 18	Mass check (<500 g +/- 50 g) + performance consistency (<10% drop). Ensure scale unit is consistent over rounds and ensure no zero-error.

Appendix B

Mouse Race · Choice Matrices · EE22005 Group 18 · yamae20, msmk23, mua35, xm482, sk3307, jh4053

SP1 · N1 - Drive & chassis configuration

Target: Lap time < 25 s (20 s +/- 5 s)

Criteria	Weight	4-Wheel chassis	2-Wheel + skid	Ball caster
Speed / lap time	0.40	3	5	3
Friction reduction	0.30	2	5	4
Stability	0.20	5	4	3
Simplicity	0.10	3	5	3
Total	1.00	3.1	4.8	3.3
Chosen: 2-Wheel drive with rear low-friction skid				

SP2 · N2 - Signal detection circuit

Target: Signal detection >90% reliability

Criteria	Weight	Simple diode rect.	Direct AC processing	Precision rect.
Detection accuracy	0.45	2	3	5
Signal clarity	0.25	2	3	5
Noise reduction	0.20	2	2	5
Implementation	0.10	5	4	3
Total	1.00	2.3	2.9	4.8
Chosen: Precision rectifier + peak detector				

SP3 · N3 - Motor driver type

Target: Speed drop <10% over 5 runs

Criteria	Weight	Linear transistor	MOSFET switching	Direct battery
Efficiency	0.40	2	5	3
Thermal performance	0.25	2	5	4
Speed control range	0.20	3	5	1
Ease of implementation	0.15	4	3	5
Total	1.00	2.5	4.7	3.2
Chosen: MOSFET switching motor driver				

SP4 · N4 - Speed control method

Target: RPM variation <10% during turns

Criteria	Weight	Fixed voltage	PWM control	Manual resistance
Control precision	0.40	2	5	2
Energy efficiency	0.25	3	5	2
Responsiveness	0.20	2	5	2
Implementation	0.15	5	3	4
Total	1.00	2.7	4.7	2.3
Chosen: PWM (pulse width modulation) control				

SP5 · N5 - Coil mounting method

Target: Adjustment completed in <2 min without rebuild

Criteria	Weight	Soldered to PCB	Screw/bracket mount	Tape + croc clips
Speed of adjustment	0.40	1	3	5
Flexibility	0.30	1	3	5
No permanent mod.	0.20	1	4	5
Simplicity	0.10	3	2	5
Total	1.00	1.2	3.1	5.0
Chosen: Temporary fixing using sellotape + crocodile clips				

SP6 · N6 - Component sourcing strategy

Target: Total system cost <£30 +/- £5

Criteria	Weight	Custom PCB design	Off-the-shelf	Specialist parts
Cost	0.40	1	5	2
Availability	0.30	2	5	2
Performance	0.20	5	3	5
Ease of integration	0.10	2	5	2
Total	1.00	2.2	4.6	2.6
Chosen: Off-the-shelf lab components				

SP7 · N7 - Soldering fume control

Target: >30 cm clearance, fumes visibly displaced

Criteria	Weight	Respirator mask	Extraction fan	Local fan
Effectiveness	0.40	3	5	5

SP8 · N8 - Component reuse strategy

Target: >80% components reused across builds

Criteria	Weight	Single-use design	Fully modular	Hybrid reuse
Sustainability (% reuse)	0.40	1	5	4

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Practicality	0.30	2	3	5
Availability	0.20	3	2	5
Comfort	0.10	2	4	5
Total	1.00	2.6	3.7	5.0

Chosen: Use of local fan to direct fumes away from user

Practicality	0.30	4	2	4
Time efficiency	0.20	4	2	4
Performance retention	0.10	5	3	4
Total	1.00	2.9	3.3	4.0

Chosen: Hybrid reuse approach (reuse key parts, rebuild minor sections)

SP9 · N9 - Wiring layout approach

Target: No loose wires >3 cm, clear routing visible

Criteria	Weight	Loose wiring	Cable-tied routing	Minimise wire length
Visual clarity	0.35	1	4	5
Ease of debugging	0.30	2	3	5
Aesthetic quality	0.20	1	3	5
Accessibility	0.15	5	3	5
Total	1.00	1.9	3.4	5.0

Chosen: Reduce and remove unnecessary wire length

SP10 · N10 - Documentation approach

Target: All members operable after <10 min explanation

Criteria	Weight	Undocumented system	Basic labels	Full docs + GitHub
Clarity	0.40	1	3	5
Ease of use	0.30	1	3	5
Team inclusivity	0.20	1	2	5
Time efficiency	0.10	5	4	3
Total	1.00	1.4	2.9	4.8

Chosen: Fully documented and labelled system on GitHub and Teams

SP11 · N11 - Electrical safety approach

Target: Operate at <6 V DC, no exposed conductors

Criteria	Weight	No compliance	Basic safe design	Standards-informed
Safety	0.40	1	3	5
Compliance	0.30	1	2	5
Professional practice	0.20	1	3	5
Documentation	0.10	1	2	5
Total	1.00	1.0	2.6	5.0

Chosen: Standards-informed design (BS EN 61010-1, RoHS, WEEE)

SP12 · N12 - Team meeting structure

Target: Each member >1 idea per meeting, no one >50% input

Criteria	Weight	Free conversation	Rotating turns	Team leader control
Inclusivity	0.40	2	5	1
Fair participation	0.30	2	5	2
Idea diversity	0.20	3	5	2
Ease of use	0.10	5	4	3
Total	1.00	2.5	4.9	1.7

Chosen: Rotating speaking turns during discussions

SP13 · N13 - Skid material source

Target: >80% skid material reused, <20% new plastic

Criteria	Weight	New acrylic sheet	Prev. year skids	Workshop offcut
Waste reduction	0.45	1	5	4
Material availability	0.25	4	2	5
Environmental impact	0.20	1	4	5
Cost	0.10	2	4	5
Total	1.00	1.9	4.0	4.6
Chosen: Use scrap/offcut plastic from workshop stock				

SP14 · N14 - Skid fabrication method

Target: 0 incidents, controlled cutting (no uncontrolled fractures)

Criteria	Weight	Clamp + saw	Score + snap	Laser cutter
Safety	0.40	5	2	4
Control / precision	0.30	5	2	5
Availability	0.20	5	5	2
Material suitability	0.10	5	2	4
Total	1.00	5.0	2.6	3.9
Chosen: Use clamp and saws for controlled cutting				

SP15 · N15 - Weight reduction strategy

Target: Total robot mass <500 g +/- 50 g, <10% speed drop

Criteria	Weight	Compact integrated	Lighter skid material	Wireless power
Weight reduction	0.40	5	4	3
Practicality	0.30	5	5	1
Reliability	0.20	5	5	2
Implementation feasib.	0.10	5	5	1
Total	1.00	5.0	4.6	2.0
Chosen: Compact and tightly integrated design				

Score scale: 1 = Poor 2 = Below average 3 = Average 4 = Good 5 = Excellent | Green = winner (highest weighted total) | Total = product of weight and score